

Question and Dataset

1 Assignment

For the group project, your team (groups of four) must pick an organization, a managerial question of interest, and an *obtainable* dataset that can be used to help provide insight into the question. The dataset must have at least 500 rows. Some examples are (but not limited to):

Organization	Question	Dataset
Lyft	How “dynamically” is Uber pricing with respect to weather? Is it more dynamic than ours (Lyft’s)?	Kaggle “Uber & Lyft Cab Prices.” Contains ~693K real-time price quotes in Boston collected every 5 minutes over 22 days, merged with weather data (temperature, rain, wind).
A real estate development firm	Which U.S. metro areas should the firm prioritize for new residential investment, based on where home values are appreciating fastest relative to available inventory?	Zillow Research Data (zillow.com/research/data). Monthly time-series of home values, rental prices, inventory counts, and market heat indices by metro, city, and zip code.
Universal Music Group	What audio features (danceability, energy, tempo, etc.) separate songs that chart from songs that don’t, and how have these characteristics changed over time?	Kaggle “Spotify Tracks Dataset.” Contains 114K tracks with numeric audio features (danceability, energy, tempo, valence, acousticness) and popularity scores across 125 genres.
City of Chicago Dept. of Public Health	Where should the city focus its limited food-inspection resources to catch the most violations?	Kaggle “Chicago Food Inspections.” Contains 200K+ inspection records with establishment type, violation descriptions, pass/fail results, and geographic coordinates.

These are suggestions to get you started. You are welcome (and encouraged!) to find your own organization and dataset. The company or organization should exist in reality. As a group, you should identify (1) an organization, (2) a managerial research question and whether it is causal or descriptive, (3) a dataset that can help answer it, and (4) the key components of your question type (see Deliverables). See the Deliverables section below for details.

If you're struggling with the open-endedness of the assignment, pick one of the examples above and make it your own. Additional datasets and dataset search engines are listed at the end of this document.

Causal vs. descriptive questions

Your research question does not have to be causal. A descriptive or predictive question is perfectly fine, as long as it is rich enough to sustain an interesting analysis. Below is what each type of question requires.

Causal questions

The goal of a causal analysis is to estimate the effect of some policy, regulation, business decision, or event on a measurable outcome (profit, revenue, sales, waste, traffic, crime, health outcomes, etc.). In business analytics we borrow the language of medicine: the policy or event is the “treatment,” the affected units are “treated,” and the unaffected units are the “control group.”

We will cover four *identification strategies* in this course. An identification strategy is a research design that lets you credibly isolate the causal effect of a treatment on an outcome. You do not need to pre-commit to one now; we will learn later which strategy is appropriate for which settings. The key distinction between them is the structure of your data: how many treated units you have, how many control units, and what kind of variation identifies the effect.

Difference-in-Differences (DID). The most common and most accessible strategy. You need: (1) a treatment that switches on at a specific time for some units but not others, (2) treated units and control units, and (3) data before and after the treatment for both groups. Works with any number of treated and control units. Examples:

- Under the Affordable Care Act, some states expanded Medicaid eligibility while others refused. Treated: expansion states. Control: non-expansion states. Outcome: uninsured rates, hospital finances.
- Several U.S. states legalized recreational marijuana starting in 2012, while others have not. Outcome: crime rates (FBI data), tax revenue, property values (Zillow).
- A company rolls out a new pricing algorithm in 20 of its 100 stores. Treated: the 20 stores. Control: the 80 stores that kept the old algorithm. Outcome: revenue per store.

Synthetic Control. Best when you have a single treated unit or a handful of treated units, a clear treatment date, and a large number of control units relative to the treated units. The method constructs a “synthetic” version of the treated unit from a weighted combination of

control units to serve as the counterfactual. You need: (1) one or a small number of treated units, (2) a clear treatment date, and (3) a large donor pool of comparable untreated units with pre-treatment data. Examples:

- Several U.S. states passed laws banning commercial food waste from landfills (5 treated states) while the remaining ~31 states did not. The method builds a “synthetic” version of each treated state from the control states to estimate what disposal would have looked like without the ban. See Anglou, Sanders, and Stamatopoulos (2025, *Science*).
- What did California’s cap-and-trade program do to the state’s carbon emissions? California is the single treated unit. The donor pool is the other 49 states that did not adopt cap-and-trade.

Regression Discontinuity (RDD). A powerful strategy, but harder to find good settings for. You need: (1) a sharp cutoff or threshold that determines who gets treated, and (2) units on either side of the cutoff that are comparable. The key requirement is that whether a unit falls just above or just below the cutoff is quasi-random and not easily “gamed.” Examples:

- Baseball games decided by 1 run. A team that wins by 1 run versus loses by 1 run experienced nearly identical games, but the outcome (win vs. loss) flips. This lets you estimate the causal effect of a win on subsequent attendance.
- Medicare eligibility at age 65. People who are 64 and 66 are nearly identical, but one group has access to Medicare and the other does not. You cannot game your age. Does gaining Medicare reduce mortality? Hospitalization data and mortality statistics are publicly available through CMS and the CDC.

A/B Test (Randomized Controlled Trial). The gold standard of causal inference: units are randomly assigned to treatment and control. Because assignment is random, you only need data from a single point in time (no before/after comparison needed). In a business context, these are typically called A/B tests. You need: (1) random assignment of units to treatment and control, and (2) an outcome measured for both groups. The challenge is that public A/B test data is much harder to find than observational data, since most companies keep their experiments proprietary. Example:

- An e-commerce retailer randomly splits 64,000 past customers into three groups: one receives an email campaign featuring men’s merchandise, one receives women’s merchandise, and one receives no email. The outcome is whether the customer visited the website, made a purchase, and how much they spent. This dataset (the Hillstrom Email Marketing Experiment) is publicly available as a free CSV download.

Summary:

	DID	Synthetic Control	RDD	A/B Test
Treated units	Any number	One or a handful	Units just above/below cutoff	Randomly assigned to treatment
Control units	Any number	Large number relative to treated	Units on other side of cutoff	Randomly assigned to control
Key requirement	Pre- and post-treatment data for both groups	Large donor pool of comparable untreated units	Cutoff is quasi-random, not gameable	Random assignment (rare in public data)

The focus of a causal project should be on credibly nailing the measurement and providing convincing evidence that the analysis is correct and robust. It does not matter what the treatment effect turns out to be or whether the policy “worked.” What matters is that the analysis is sound. A well-executed null result is more impressive than a sloppy positive one.

If you are drawn to a causal question, do not be intimidated; these strategies are more accessible than they sound. Come to office hours if you want to talk through whether your question is feasible and which strategy fits!

Descriptive questions

A descriptive question asks: what is happening in the data, and why does it matter? There is no treatment, no control group, and no causal claim. The goal is to characterize patterns, relationships, or trends in a way that informs a business decision.

A descriptive question should have:

1. A clear phenomenon to characterize. What pattern, relationship, or trend are you trying to understand? Examples:
 - How do Airbnb listing prices vary across San Diego neighborhoods, and what features drive the highest premiums?
 - What audio features distinguish songs that chart on Spotify from songs that do not?
 - Which U.S. metro areas have the fastest-growing home values relative to available inventory?
2. Multiple dimensions of variation. A good descriptive analysis slices the data in several meaningful ways: across time, geography, customer segments, product categories, or other groupings. The richer the variation you uncover, the more interesting the project.
3. Sub-questions that build a narrative. Each sub-question should reveal something new and build on the previous one. The final product should tell a coherent story, not present a grab-bag of unrelated charts.
4. Actionable takeaways. Because there is no causal claim to anchor the analysis, the

value comes from surfacing patterns that management can act on. “Listings with pools command a 22% premium in coastal zip codes but only 6% inland” is more useful than “here is a histogram of prices.”

Because descriptive questions carry less methodological burden (you do not need to defend a quasi-experiment), the bar for depth and richness is higher. If you choose a descriptive project, plan to go deeper: more sub-questions, more angles, more polished visualizations, and a more complete narrative than you would need for a causal project where the identification strategy is the star of the show.

2 Deliverables

Submit a short document that answers:

1. What is the organization?
2. Research question/classification: What is the single, definitive managerial research question your organization is facing? Is it causal or descriptive? (See Section 1 for what each requires.)
3. What is your dataset? Where did you get it, and what does it contain? Please use AI to create a table describing the scope of the data: the number of observations, the dates covered, the unit of observation (i.e., what each row represents), and what the columns/fields are. See the example solution below.
4. Address the key components of your question type:
 - *If causal:* Which method (DID, synthetic control, RDD, or A/B test)? What is the treatment? What are the treated units, and how many are there in your data? What is the control group (or cutoff, or donor pool), and how many control units are there? What is the outcome you are measuring?
 - *If descriptive:* What are the main dimensions of variation you plan to explore (e.g., time, geography, customer segments)? What are two or three sub-questions that build toward answering the broader research question? What actionable takeaways do you expect to surface for management?

The intended audience of this document should be the management of the organization for whom you are conducting the study. It should specify what the research question is, what data you will use, and how you plan to use it.

Please submit your writeup on Canvas. The broad scope of this project is designed to offer you a large degree of freedom, so I ask that you do not misuse it. I suggest choosing a classroom-appropriate topic that is able to capture the interest of your group and your peers.

3 Example Solutions

Causal example (DID)

1. **Organization:** A state legislature or the U.S. Department of Justice, interested in the public safety effects of marijuana legalization.
2. **Research question/classification:** Did legalizing recreational marijuana increase or decrease crime rates? This is a causal question using difference-in-differences.
3. **Dataset:** FBI crime data (see Section 4) combined with state marijuana legalization dates. The easiest source is the CORGIS State Crime dataset (corgis-edu.github.io/corgis/csv/state_crime/, free CSV download), containing 3,115 observations covering 50 states plus D.C. spanning 1960 to 2019. Each row is a state-year. Key fields:

Field	Description
State	State name
Year	Calendar year
Data.Population	State population
Data.Rates.Violent.All	Violent crime rate per 100K
Data.Rates.Violent.Murder	Homicide rate per 100K
Data.Rates.Property.All	Property crime rate per 100K
Data.Totals.Violent.All	Total violent crime count
Data.Totals.Property.All	Total property crime count

4. **Key components:** Method is DID. The treatment is legalization of recreational marijuana. Treated units are the states that legalized (starting with Colorado and Washington in 2012, with more states following in subsequent years). Control units are the states that have not legalized. The outcome is crime rates (violent crime, property crime, drug offenses, etc.). We have data before and after legalization for both groups.

Causal example (Synthetic Control)

1. **Organization:** U.S. Environmental Protection Agency (or a state environmental agency).
2. **Research question/classification:** Did state-level food waste disposal bans reduce the amount of waste sent to landfills? This is a causal question using synthetic control.
3. **Dataset:** County- and state-level waste disposal data from Anglou, Sanders, and Stamatopoulos (2025, *Science*), available from the Dryad repository (see Section 4, free

CSV download). Contains 54,580 observations of waste tonnage by disposition type across 36 U.S. states spanning 1991 to 2021, covering ~85% of the U.S. population. Each row is a county-year-type. Key fields:

Field	Description
year	Calendar year
state_id	State identifier
county_name	County name (NA if reported at state level)
type	Disposition method (disposal, composting, food_donation, etc.)
tons	Amount of waste in tons

4. **Key components:** Method is synthetic control. The treatment is the enactment of a food waste ban. Treated units are the 5 states that passed a ban. The donor pool is the ~31 states that did not. For each treated state, the method constructs a weighted combination of control states that best matches the treated state's pre-ban disposal trends, then compares actual post-ban disposal to what the synthetic version predicts. The outcome is total tons of waste disposed in landfills.

Causal example (RDD)

1. **Organization:** San Diego Padres sports franchise.
2. **Research question/classification:** How much is poor on-field performance costing the Padres in live attendance revenue? This is a causal question using regression discontinuity.
3. **Dataset:** MLB game-level attendance data from Retrosheet (retrosheet.org/gamelogs, free CSV download), containing 231,204 observations spanning 1871 to 2024. Each row is a single game. Key fields:

Field	Description
date	Date of the game
home_team	Home team name
away_team	Away team name
attendance	Number of fans in attendance
home_score	Runs scored by the home team
away_score	Runs scored by the away team
day_of_week	Day of the week

Field	Description
<code>park_id</code>	Stadium identifier

4. **Key components:** Method is RDD. The cutoff is the final score margin: games won by 1 run (margin = +1) versus games lost by 1 run (margin = -1). Games decided by a single run are quasi-random, so the outcome of these games is not easily gamed. This is especially true for higher-scoring 1-run games (e.g., 5-4 or 6-5, which are common in MLB), where both teams were clearly hitting well and neither was dominant. By comparing attendance at the next home game for narrow wins versus narrow losses, we can estimate the causal effect of a win on subsequent attendance. There are roughly 1,200 Padres home games from 2000 to 2024.

Causal example (A/B Test)

1. **Organization:** An e-commerce retailer evaluating the effectiveness of its email marketing campaigns.
2. **Research question/classification:** Do email campaigns actually drive revenue, or would these customers have purchased anyway? This is a causal question using an A/B test (randomized controlled trial).
3. **Dataset:** The Hillstrom Email Marketing Experiment (minethatdata.com, free CSV download), containing 64,000 observations. Each row is a customer. Key fields:

Field	Description
<code>recency</code>	Months since last purchase
<code>history</code>	Dollar value of past purchases
<code>mens</code>	1 if customer purchased men's merchandise in the past
<code>womens</code>	1 if customer purchased women's merchandise in the past
<code>zip_code</code>	Urban, suburban, or rural
<code>newbie</code>	1 if new customer in the past 12 months
<code>channel</code>	Purchase channel (phone, web, multichannel)
<code>segment</code>	Treatment group: Mens E-Mail, Womens E-Mail, or No E-Mail
<code>visit</code>	1 if the customer visited the website in the following 2 weeks
<code>conversion</code>	1 if the customer made a purchase
<code>spend</code>	Dollar amount spent

4. **Key components:** Method is A/B test. The treatment is receiving an email campaign (men's or women's merchandise). The control group received no email. 64,000 customers were randomly assigned to three equally sized groups. The outcomes are website visits, conversions, and spending over the 2 weeks following the campaign. Because assignment was random, any difference in outcomes between groups is attributable to the email.

Descriptive example

1. **Organization:** Airbnb (from the perspective of a host or a real estate investor evaluating short-term rental opportunities in San Diego).
2. **Research question/classification:** Where in San Diego are Airbnb hosts systematically underpricing their listings, and what listing features drive the highest premiums? This is a descriptive question.
3. **Dataset:** Inside Airbnb listing data for San Diego (insideairbnb.com/get-the-data, free CSV download), containing 11,427 listings as of the most recent scrape (February 2026). Each row is a single listing. Key fields:

Field	Description
price	Nightly price in USD
neighbourhood	San Diego neighborhood
latitude / longitude	Geographic coordinates
room_type	Entire home, private room, or shared room
accommodates	Maximum number of guests
bedrooms	Number of bedrooms
review_scores_rating	Average guest rating (0–5)
number_of_reviews	Total reviews received
amenities	List of amenities (pool, parking, AC, etc.)
availability_365	Days available in the next year

4. **Key components:** The main dimensions of variation are geography (neighborhood), listing features (room type, size, amenities), and guest reviews. Sub-questions: (a) How do prices vary across San Diego neighborhoods for comparable listings? (b) Which amenities command the largest price premiums? (c) Are there neighborhoods where listings with similar features are priced well below the market, suggesting an opportunity for hosts to raise prices? The actionable takeaway for a host or investor would be a map of where underpriced opportunities exist and which features to invest in.

4 Where to Find Datasets (and Quasi-Experiments)

For a detailed guide on finding datasets, quasi-experiments, scraping techniques, and a catalog of recommended data sources, see the companion document: [Where to Find Datasets \(and Quasi-Experiments\)](#).

5 Forbidden Question/Dataset Combinations

The following are specific question/dataset pairs where complete, copy-paste-ready analyses already exist across the internet (Kaggle notebooks, Medium articles, YouTube walkthroughs, etc.). You may use any of these datasets, but not with the canned question listed here. If you want to use one of these datasets, you must ask a genuinely original question.

1. “Predict which passengers survived the Titanic” + Kaggle Titanic dataset. Thousands of complete notebooks exist for this exact question.
2. “Predict which employees will leave the company” + IBM HR Analytics Attrition dataset. Hundreds of identical walkthroughs on Medium and Kaggle.
3. “Predict whether a telecom customer will churn” + IBM Telco Customer Churn dataset. The default churn project in every data science bootcamp.
4. “What genres does Netflix have the most of?” + Kaggle Netflix Movies and TV Shows dataset. Hundreds of identical bar-chart analyses exist.
5. “Analyze retail sales by region and product category” + Kaggle/Tableau Superstore Sales dataset. The default Tableau beginner exercise.
6. “Detect fraudulent credit card transactions” + Kaggle Credit Card Fraud dataset. Every class-imbalance tutorial uses this exact setup, and the anonymized features make every analysis look the same.
7. “Predict heart disease / classify wine quality / classify iris species / recognize handwritten digits” + the corresponding UCI or Kaggle datasets. These are textbook teaching exercises, not business projects.

The rule of thumb: if you can Google your question and dataset name together and find a complete analysis in the first page of results, you need a more original question.